



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

Introduction to Data Science

Lecture 14+15 Optimization I: Optimization Formulation + Convex Set

Outline

- Motivating Example
- Mathematical Optimization and Basic Terminologies
- Local vs. Global Optimality
- Convexity
 - Convex set
 - Convex function (next lecture)
- Example: Linear Programming

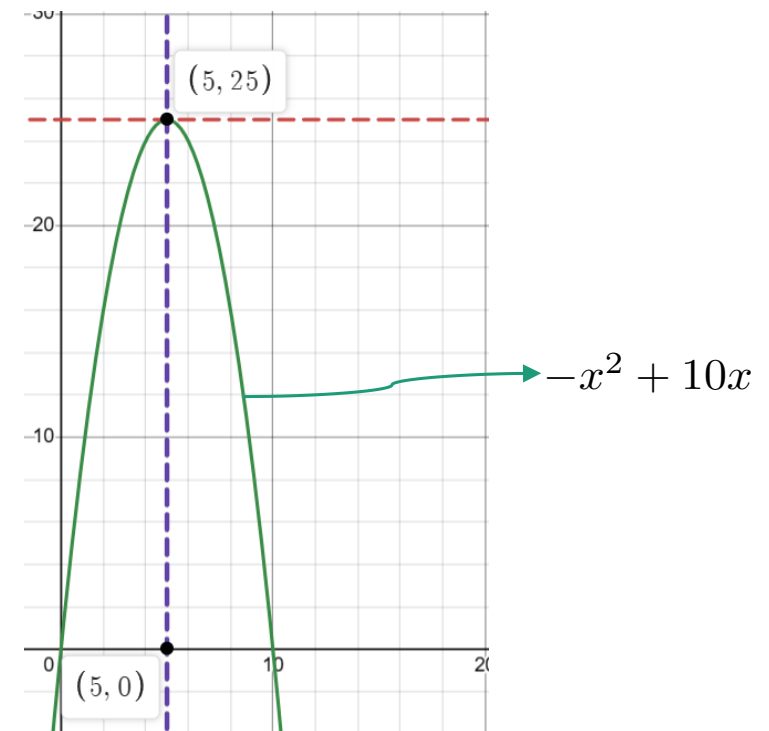
Motivating Example: Pricing with Linear Demand

- Suppose a company sells a product at a unit price x
- Demand for the product is modeled as

$$D(x) = ax + b$$

- $a < 0$: price sensitivity
 - $b > 0$: base demand
- Objective is to find a price $x \geq 0$ to maximize the revenue (price * demand)

$$f(x) = xD(x) = ax^2 + bx$$



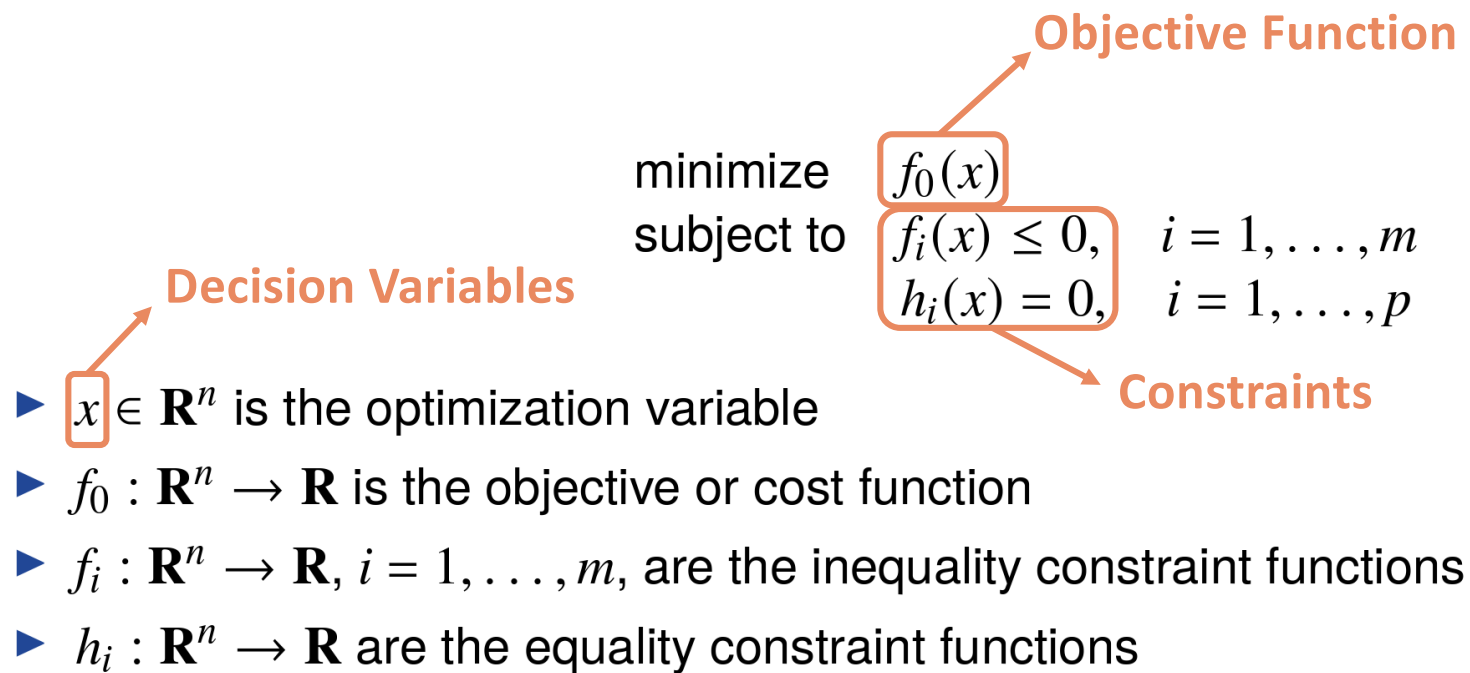
Optimization: General Form

Given: a **function** $f: A \rightarrow \mathbb{R}$ from some **set** A to the **real numbers**

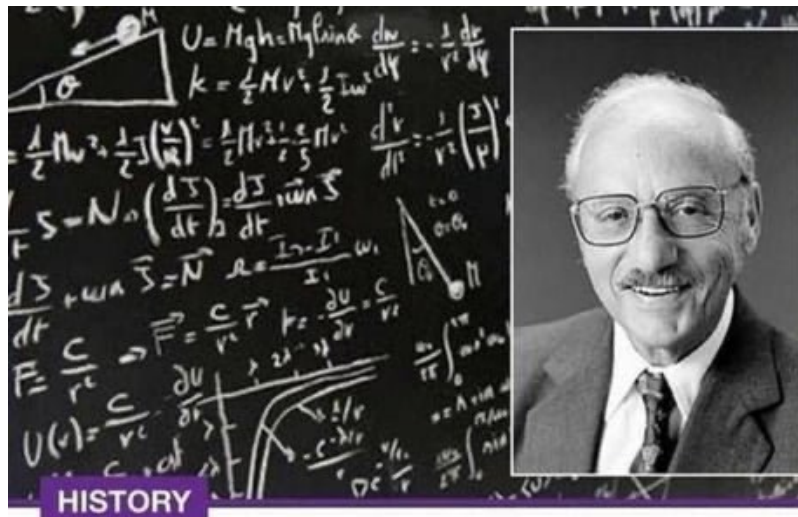
Sought: an element $\mathbf{x}_0 \in A$ such that $f(\mathbf{x}_0) \leq f(\mathbf{x})$ for all $\mathbf{x} \in A$ ("minimization") or such that $f(\mathbf{x}_0) \geq f(\mathbf{x})$ for all $\mathbf{x} \in A$ ("maximization").

- x is called the **decision variable**
 - E.g. pricing decision
- The function f is called the **objective**
 - E.g. $f(x) = ax^2 + bx$
- The set A defined by the **constraints** is called the **feasible region**
 - E.g. $x \geq 0$

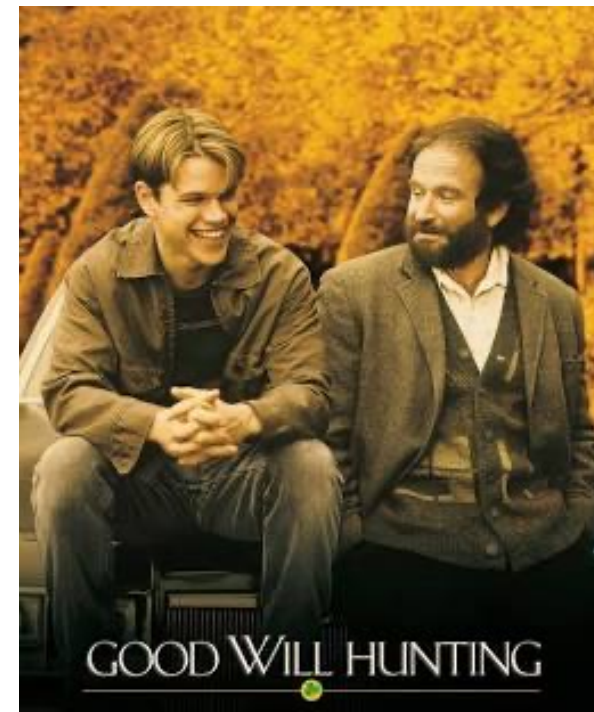
Optimization: Standard Form



Father of Math Optimization: George B. Dantzig



After getting late to class, **George Dantzig** copied from the blackboard two problems thinking they were homework, and then **solved them**. They were actually **two famous unsolved statistics problems**, which earned him his **PhD**.



Father of Math Optimization: George B. Dantzig

- George Dantzig won the **first** John von Neumann Theory Prize
- Who won the last year's John von Neumann Theory Prize?
- 2024 **Jim Dai**
- 1976 **Richard Bellman**
- 1975 **George B. Dantzig** *for his work on **linear programming***

John von Neumann Theory Prize

Awarded for	Fundamental, sustained contributions to theory in operations research and the management sciences
First award	1975
Website	John von Neumann Theory Prize 

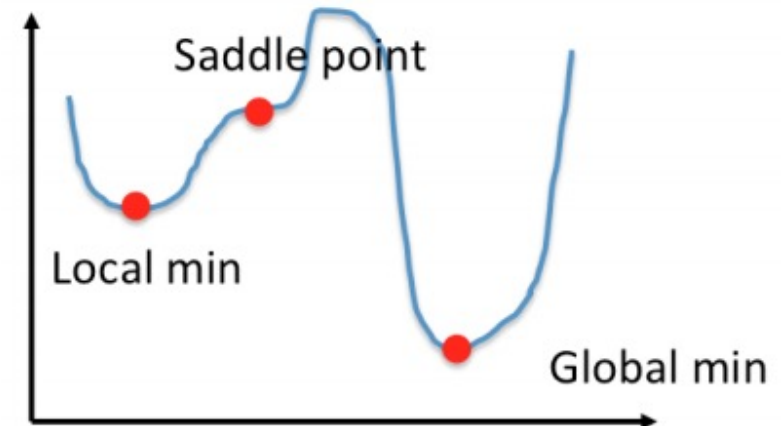
Basic Terminologies

- **Feasible Set**: the set of all points such that the *all* constraints are satisfied

- **Global optimizer**: $\mathbf{x}^*$ is a *global minimizer* if $f(\mathbf{x}^*) \leq f(\mathbf{y})$ for any $\mathbf{y}$ in the feasible set

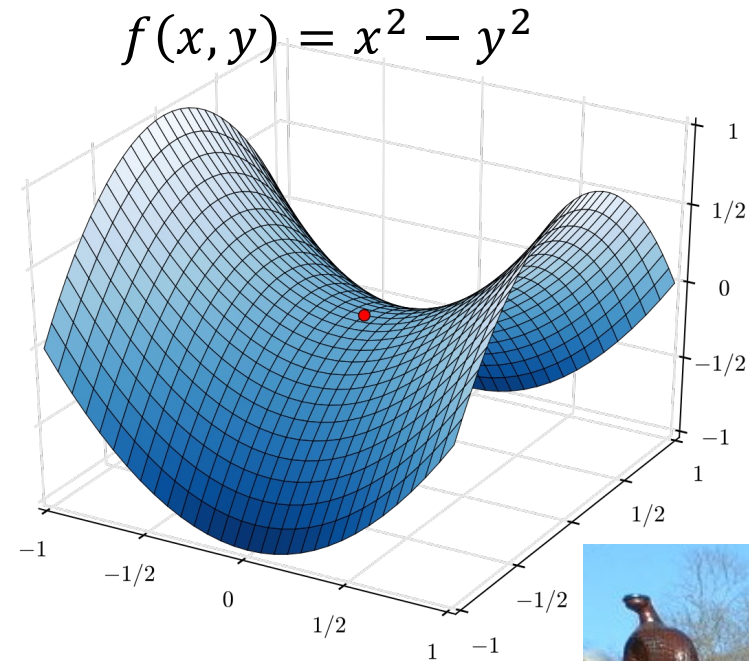
- **Local optimizer**: optimal only in its *neighborhood*

- If there exists an $\varepsilon > 0$ such that $f(\mathbf{x}) \leq f(\mathbf{y})$ for any $\mathbf{y}$ in the feasible set with $\|\mathbf{y} - \mathbf{x}\| \leq \varepsilon$, Then $\mathbf{x}$ is called a *local minimizer*.

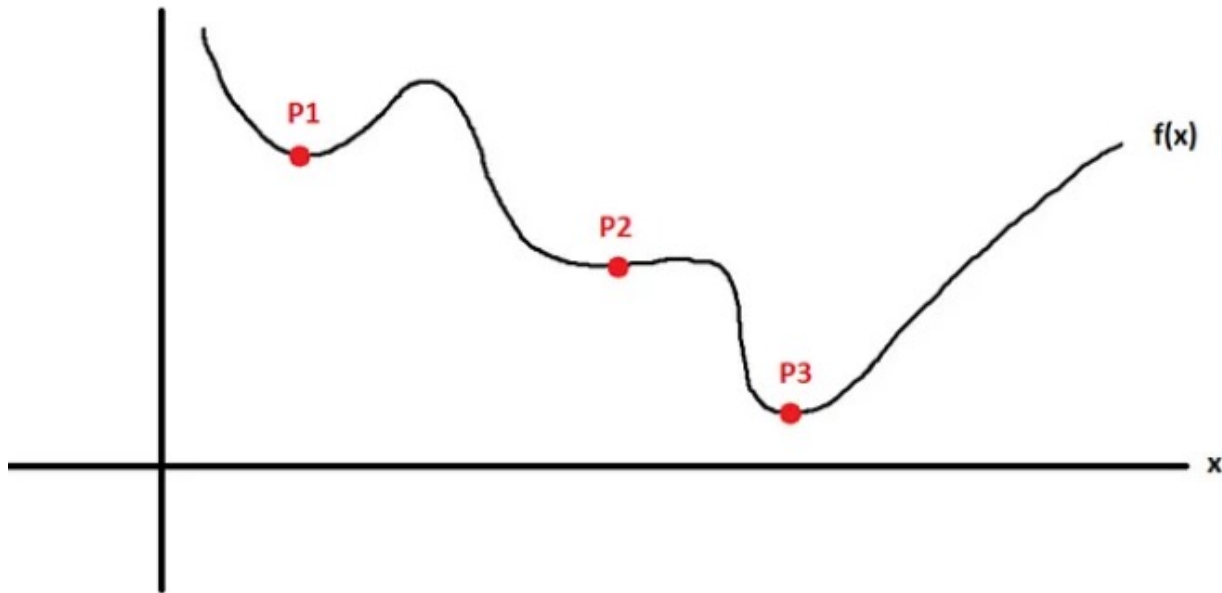


Basic Terminologies

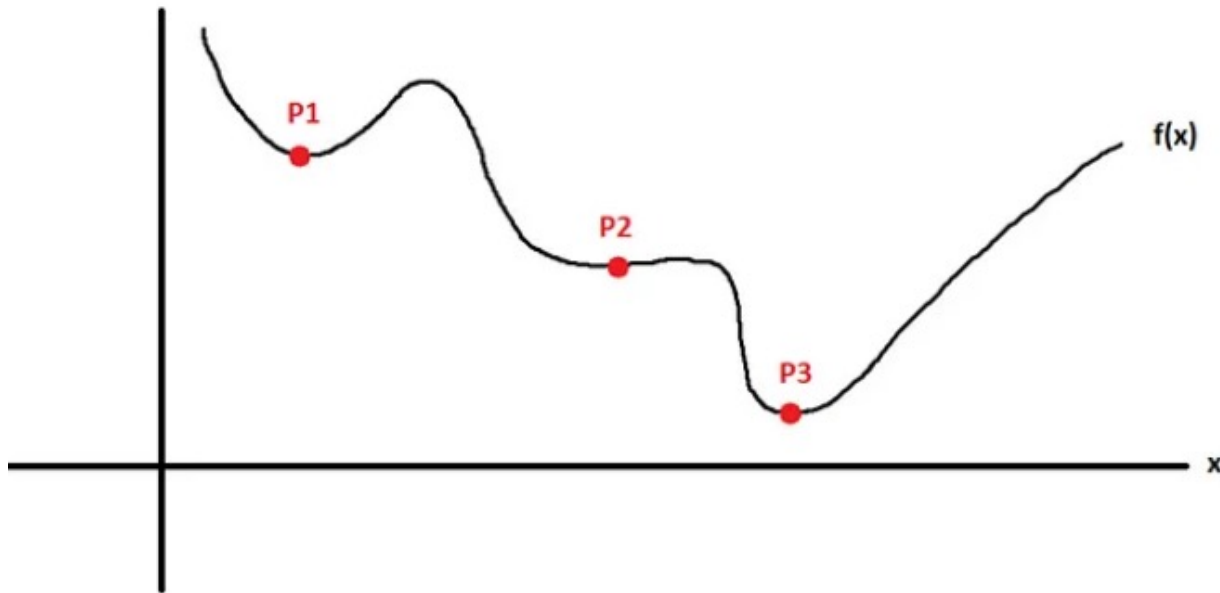
- **Saddle point**: a locally “flat” point, neither local min or max
 - The directional derivative $\left. \frac{df(\mathbf{x}^* + t\mathbf{e})}{dt} \right|_{t=0} = 0$ along any direction $\mathbf{e}$
 - **E.g.** $(0,0)$ is a saddle point of $f(x, y) = x^2 - y^2$
 - Convention: use **bold** font to represent a multi-dimension vector. A scalar is without bold



Exercise: Local or Global Minimum?



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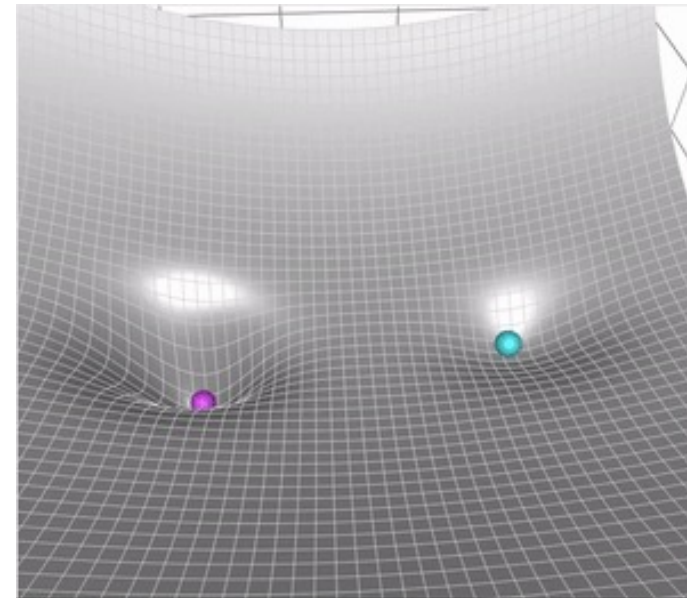
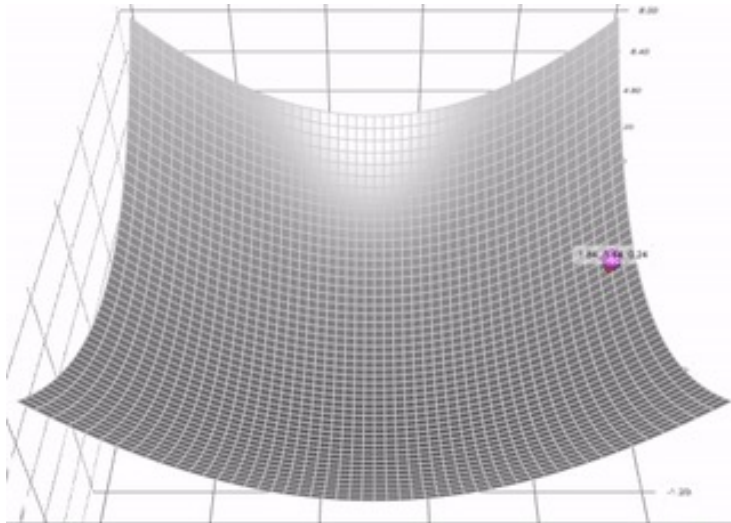


P1 is? Local Minimum

P2 is? Saddle point

P3 is? Global Minimum

Hardness of Finding Optimizers

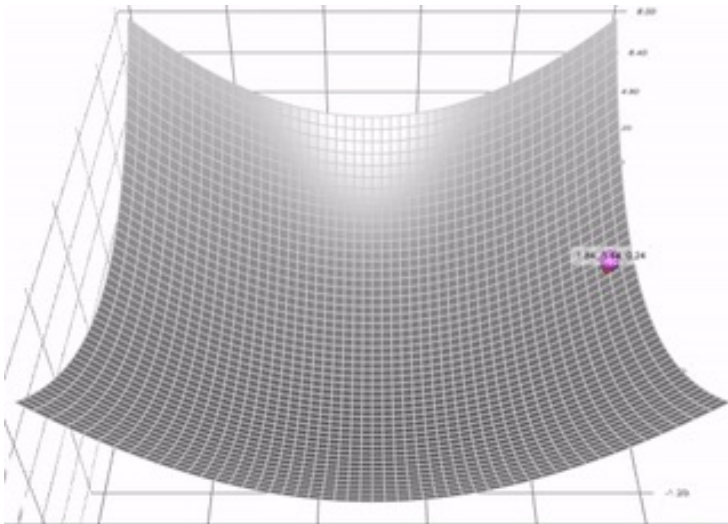


Convex Optimization

- Verifying if a point is a local minimum can be easy
- However, how can we ensure that a local minimum is indeed a global minimum?
- An important category of problems known as **convex optimization** has property that **any local minimum is also a global minimum**

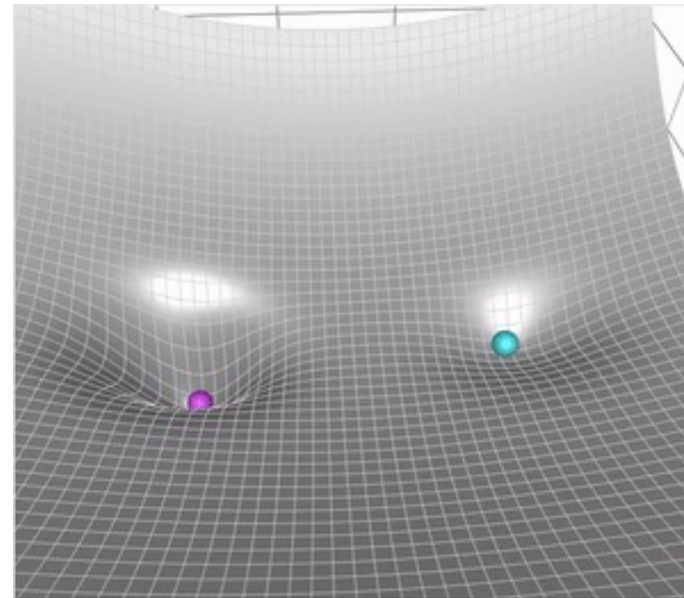
Convex vs. Non-convex

- Convex optimization



- Gradient descent finds a global optimizer efficiently

- Non-convex optimization

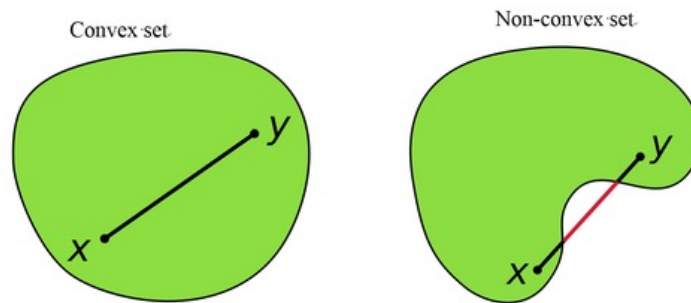


- Gradient descent tends to find only a local optimizer

How to Detect Convexity?

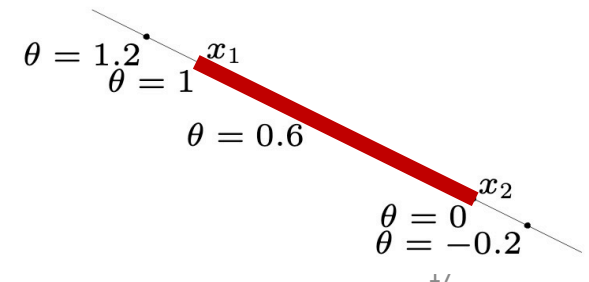
Convex Set

- A set C is a **convex set** if the *line segment* between any two points in C lies in C

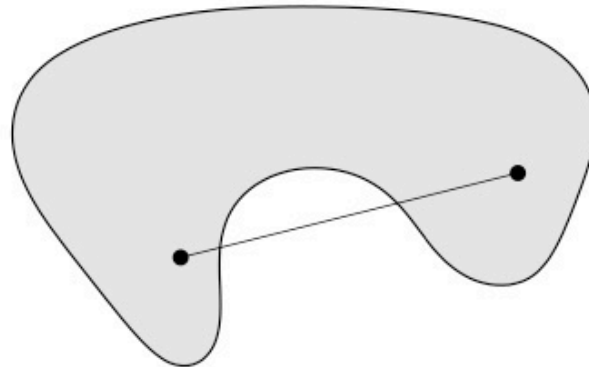
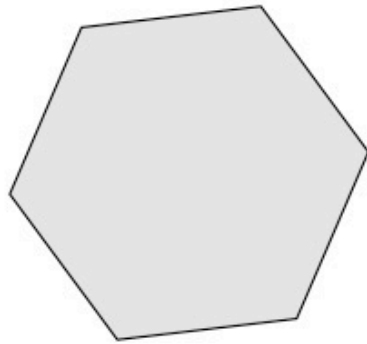


- Mathematically, a set C is convex if for any $x_1, x_2 \in C$, $\theta \in [0,1]$, it holds that

$$\theta x_1 + (1 - \theta)x_2 \in C.$$



Exercise: Which set is a convex set?



Examples of Convex Sets

- The empty set $\emptyset$, the singleton set $\{\mathbf{x}_0\}$, and the entire space R
- An interval $[a, b] \subset R$
- In R^n , the hyperplane $H := \{\mathbf{x} \in R^n : a_1x_1 + \cdots + a_nx_n = c\}$
- Half spaces, e.g., in R^2 , $H := \{(x, y) : y \leq ax + b\}$
- In R^2 , A disk with center $(0,0)$ and radius c

Rigorous Proof of Convexity of a Set

- Pick any two points in the set and show their line segment is contained in the set
- **Example:** Prove $H := \{(x, y) : y = ax + b\}$ is a convex set
- **Proof.**
 - Pick any two points $(x_1, y_1), (x_2, y_2)$ in H . By definition, we have
$$y_1 = ax_1 + b, y_2 = ax_2 + b \quad (*)$$
 - Consider any point in the line segment, represented as
$$(x_\theta, y_\theta) = \theta(x_1, y_1) + (1 - \theta)(x_2, y_2), \text{ where } 0 \leq \theta \leq 1$$
 - Verify (x_θ, y_θ) is also in H :
 - Rewrite $(x_\theta, y_\theta) = (\theta x_1 + (1 - \theta)x_2, \theta y_1 + (1 - \theta)y_2)$
 - Observe from $(*)$ that
$$\theta y_1 + (1 - \theta)y_2 = a(\theta x_1 + (1 - \theta)x_2) + b$$
 - Hence the line segment is contained in H

Exercise: Convexity of Half spaces

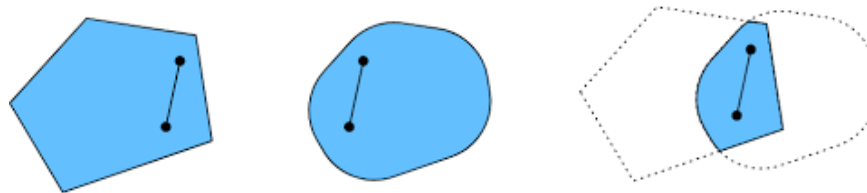
- Prove $H := \{(x, y) : y \leq ax + b\}$ is convex

Exercise: Convexity of Half spaces

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 - Hence the line segment is contained in H

Example: Intersection of Convex Sets

- If both S_1 and S_2 are convex sets, then $S_1 \cap S_2$ is also a convex set.
- Proof.
 - Pick any two points $x_1, x_2 \in S_1 \cap S_2$
 - Consider any point in the line segment, represented as
$$x_\theta = \theta x_1 + (1 - \theta)x_2, \text{ where } 0 \leq \theta \leq 1$$
 - Verify x_θ is also in $S_1 \cap S_2$:
 - Since S_1 is convex, by definition, x_θ is in S_1
 - Since S_2 is convex, by definition, x_θ is in S_2
 - Hence, x_θ is in $S_1 \cap S_2$.
 - Thus, the intersection of two convex sets is still convex

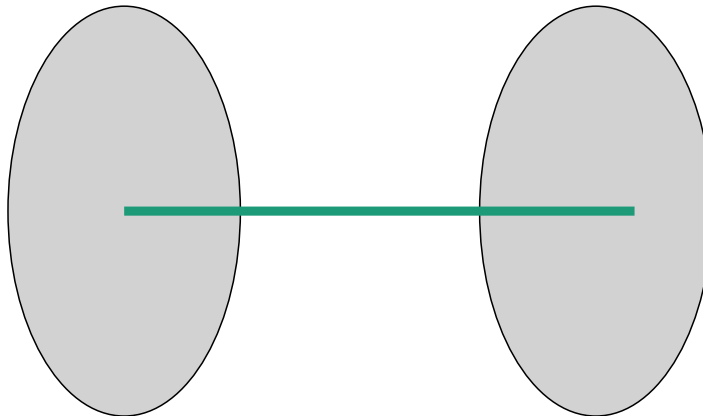


Exercise: Union of Convex Sets

- If both S_1 and S_2 are convex sets, is their union $S_1 \cup S_2$ also convex?

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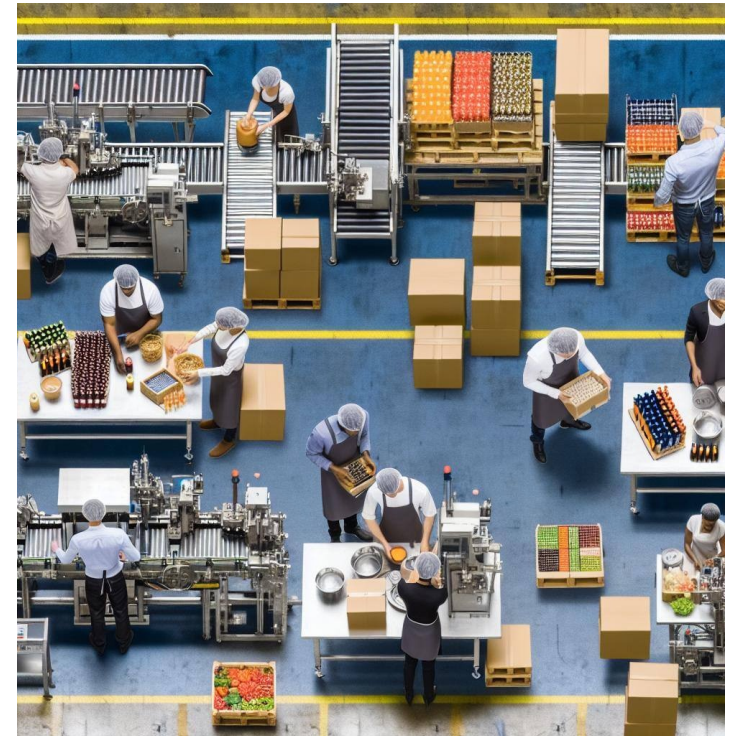


Example: Production Planning

A factory produces two types of divisible products, **A** and **B**, using two resources: **labor hours** and **raw materials**.

	Product A	Product B	Availability
Labor hours	2	1	100
Raw materials	3	2	120

The unit profit for products A and B are \$40 and \$30, respectively. The factory wants to **maximize profit** while considering resource constraints.



Decisions and Objective

Decision Variables:

- x_A : the number of **Product A** units produced
- x_B : the number of **Product B** units produced.

Objective:

The goal is to **maximize total profit**:

$$\max \quad 40x_A + 30x_B$$

Constraints

	Product A	Product B	Availability
Labor hours	2	1	100
Raw materials	3	2	120

1. Labor Hours:

- Each unit of **Product A** requires **2 hours** of labor.
- Each unit of **Product B** requires **1 hour** of labor.
- The factory has **a maximum of 100 labor hours available**.

$$2x_A + x_B \leq 100$$

Constraints

	Product A	Product B	Availability
Labor hours	2	1	100
Raw materials	3	2	120

2. Raw Materials:

- Each unit of **Product A** requires **3 kg** of raw material.
- Each unit of **Product B** requires **2 kg** of raw material.
- The factory has **a maximum of 120 kg of raw material available**.

Constraints

	Product A	Product B	Availability
Labor hours	2	1	100
Raw materials	3	2	120

2. Raw Materials:

- Each unit of **Product A** requires **3 kg** of raw material.
- Each unit of **Product B** requires **2 kg** of raw material.
- The factory has **a maximum of 120 kg of raw material available**.

$$3x_A + 2x_B \leq 120$$

Constraints

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- Each unit of **Product A** requires **3 kg** of raw material.
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Any other constraints?

Constraints

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- Each unit of **Product A** requires **3 kg** of raw material.
- Each unit of **Product B** requires **2 kg** of raw material.
- The factory has **a maximum of 120 kg of raw material available**.

$$3x_A + 2x_B \leq 120$$

Any other constraints? Yes, non-negativity $x_A \geq 0, x_B \geq 0$

Linear Program

Three elements: decision variables, objective, constraints

$$\begin{array}{ll}\max & 40x_A + 30x_B \\ \text{s.t.} & 2x_A + x_B \leq 100 \\ & 3x_A + 2x_B \leq 120 \\ & x_A \geq 0, x_B \geq 0\end{array}$$

- s.t. stands for *subject to*
- The objective function and all constraints are *linear* functions of the decision variables. This type of problems is called a **linear program**.

Feasible Region of a Linear Program

- The **feasible region** is the set of all points that satisfy all constraints simultaneously.
- In the previous example, it is

$$S = \{(x_A, x_B) \mid 2x_A + x_B \leq 100, \quad 3x_A + 2x_B \leq 120, \quad x_A \geq 0, \quad x_B \geq 0\}$$

Exercise: Convexity of Feasible Region of a Linear Program

- Using results from the previous lecture, prove the following feasible region is convex

$$S = \{(x_A, x_B) \mid 2x_A + x_B \leq 100, \quad 3x_A + 2x_B \leq 120, \quad x_A \geq 0, \quad x_B \geq 0\}$$

Exercise: Convexity of Feasible Region of a Linear Program

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- Proof:
 - We have shown that (i) A half space is convex, and (ii) the intersection of convex sets are still convex.
 - Since S is the intersection of half spaces, it is convex.
- Linear programs are convex (we have shown the feasible set is convex, and next lecture we will show the objective is a convex function).

Summary

- Three elements in an optimization problem
 - Decision, objective, constraints
- Optimality
 - Global optimal, local optimal, saddle point
- Convex set
 - Common convex sets: hyperplane, half space, feasible regions of a linear program
 - Operations that preserve convexity: intersection (but not union)